

Technical Description

Technical guideline

Gas system emissions

Series 400

Series 4000

MS61026/00E

SLU Name	Sales Name	
GG06R0400A1	MTU 6R400 GS	
GG06R0400B1	MTU 6R400 GS	
GG06R0400C1	MTU 6R400 GS	
GG12V0400A1	MTU 12V400 GS	
GG12V0400C1	MTU 12V400 GS	
GG08V4000A1	MTU 8V4000 GS	
GG08V4000A2	MTU 8V4000 GS	
GG12V4000A1	MTU 12V4000 GS	
GG12V4000A2	MTU 12V4000 GS	
GG16V4000A1	MTU 16V4000 GS	
GG16V4000A2	MTU 16V4000 GS	
GG20V4000A1	MTU 20V4000 GS	
GG20V4000A2	MTU 20V4000 GS	

Table 1: Applicability

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Table of Contents

1 Emission		1.7 Sources and Literature	17
1.1 Preface	4		
1.2 Emission limits	6	2 Appendix A	
1.2.1 Raw measured values	6	2.1 Conversion tables	18
1.2.2 Emissions with reference values	8	2.2 Material values of gases	23
1.2.3 Specific emissions	9	2.3 Abbreviations	24
1.3 Gas engine emissions	10	2.4 MTU Onsite Energy contact person / service partner	25
1.4 Emission measurements during commissioning	11		
1.5 Emission measurement during maintenance work	15	3 Appendix B	
1.6 Flow chart and example of emission setting	16	3.1 Index	26

1 Emission

1.1 Preface

The worldwide delivery of MTU gas gensets for stationary applications is associated with more stringent requirements with regard to the observance of country-specific emission guidelines. Prior to initial operation of the genset, local regulations of the respectively responsible approving authorities must be obtained and strictly observed.

Details on project-specific emissions can be found in the technical data (TD) of the customer documentation.

In case of doubt during commissioning / service work, project management (MTU) and the owner / operating company must be notified of this regulations immediately. The necessary information must be obtained from the responsible offices and documented.

Special features in EPA area:

In the EPA area, the stationary gas engines for power generation are currently certified / approved together with the complete plant. The emission measurement (Abnahmelauf = „Emission Compliance Test“) is carried out by an independent („Third Party“) test institute. The basic principles of emission legislation for stationary external-ignition combustion engines is described in the valid respective version of the 40 CFR Part 60, Subpart JJJJ .

NOTE:

The emissions to be observed may vary according to the fuel type (natural gas, biogas, etc.), type of application (standby power, continuous power, etc.), lean-burn concept (lean engine, lambda-1 engine, etc.) power output, plant type and year of manufacture.

In addition, the emissions are specified in different units – it is particularly important to note this!

The 40 CFR Part 60, Subpart JJJJ forms the basis of the statutory emission requirements in the EPA area for gas engines, but also refers to other laws or specifications. In the EPA area, it is also possible that additional local approving authorities pass more stringent or additional limit values to those contained in the 40 CFR Part 60, Subpart JJJJ.

For MTU Series 4000 gas engines in application class 3a, the following limit values apply as per 40 CFR Part 60, Subpart JJJJ: (Status 28. June 2011).

Table 1 to Subpart JJJJ of Part 60-NO_x, CO, and VOC Emission Standards for Stationary Non-Emergency SI Engines ≥ 100 HP (Except Gasoline and Rich Burn LPG), Stationary SI Landfill / Digester Gas Engines, and Stationary Emergency Engines > 25 HPHP

			Emission Standards ^a					
			g/HP-hr			ppmvd at 15 % O ₂		
Engine type and fuel	Maximum engine power	Manufacture date	NO _x	CO	VOC _d	NO _x	CO	VOC _d
Non-Emergency SI Natural Gas ^b and Non-Emergency SI Lean Burn LPG ^b	100 ≤ HP < 500	7/1/2008	2.0	4.0	1.0	160	540	86
		1/1/2011	1.0	2.0	0.7	82	270	60
Non-Emergency SI Lean Burn Natural Gas LPG	500 ≤ HP < 1.350	1/1/2008	2.0	4.0	1.0	160	540	86
		7/1/2010	1.0	2.0	0.7	82	270	60

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Non-Emergency SI Natural Gas and Non-Emergency SI Lean Burn LPG (except lean burn $500 \leq \text{HP} < 1.350$)	HP ≥ 500	7/1/2007	2.0	4.0	1.0	160	540	86
	HP ≥ 500	7/1/2010	1.0	2.0	0.7	82	270	60
Landfill / Digester Gas (except lean burn $500 \leq \text{HP} < 1.350$)	HP < 500	7/1/2008	3.0	5.0	1.0	220	610	80
		1/1/2011	2.0	5.0	1.0	150	610	80
	HP ≥ 500	7/1/2007	3.0	5.0	1.0	220	610	80
		7/1/2010	2.0	5.0	1.0	150	610	80
Landfill / Digester Gas Lean Burn	$500 \leq \text{HP} < 1.350$	1/1/2008	3.0	5.0	1.0	220	610	80
		7/1/2010	2.0	5.0	1.0	150	610	80
Emergency	$25 < \text{HP} < 130$	1/1/2009	^c 10	387	N/A	N/A	N/A	N/A
	HP ≥ 130		2.0	4.0	1.0	160	540	86
^a Owners and operators of stationary non-certified SI engines may choose to comply with the Emission Standards in units of either g/HP-hr or ppmvd at 15 percent O₂. ^b Owners and operators of new or reconstruct non-emergency lean burn SI stationary engines with a site rating of greater than or equal to 250 brake HP located at a major source that are meeting the requirements of 40 CFR part 63, subpart ZZZZ, Table 2a do not have to comply with the CO Emission Standards of Table 1 of this subpart. ^c The Emission Standards applicable to emergency engines between 25 HP and 130 HP are in terms of NO_x + HC. ^d For purposes of this subpart, when calculating emissions of volatile organic compounds, emissions of formaldehyde should not be included. [76 FR 37975, June 28, 2011] Quelle: http://www.ecfr.gov								

These definitions are taken from the current legislation as per 40 CFR Part 60, Subpart JJJJ and correspond to the current status as per the change history.

As the legislation is subject to constant change, in case of doubt the status updated daily on the EPA - Homepage shall apply (cf. 1.7. Sources and Literature).

1.2 Emission limits

1.2.1 Raw measured values

Depending on the emission measuring instrument, the measured emissions are determined volumetrically or gravimetrically (mass-based). The exhaust gas measuring kits used at MTU usually record the raw measured values volumetrically. The raw measured values are usually expressed in parts per million (ppm) or percent by volume (Vol %). The conversion from ppm to Vol % is made according to the following specification:

$$1 \text{ Vol \%} = 10\,000 \text{ ppm or } 1 \text{ ppm} = 0.0001 \text{ Vol \% [1]}$$

Raw measured values in ppm are output, e.g. by the TESTO exhaust gas measurement kits (NO_x, O₂ and CO) as well as by the TESTA FID (flame ionization detectors). Similarly, some data is specified on the test report from the factory acceptance test in Vol. % or ppm.

Example of factory acceptance sequence

1 Acceptance Test Record

2 Exhaust emissions

O2 tr.		NOx tr.		NOx tr. 5% O2		CO tr.		CO tr 5% O2		THC	
Vol%	ppm	ppm	mg/nm ³	ppm	mg/nm ³	ppm	mg/nm ³	ppm	mg/nm ³	ppm	mg/nm ³

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NOTE:

The type of measured data acquisition must always also be specified (moist or dry) – the measured values differ, in part, significantly! It may be necessary to make a moist-dry conversion!

The conversion from volumetrically recorded measured variable to a gravimetric measured variable is carried out with the help of the molar masses and molar volumes of the respective exhaust gas component.

$$c_{\text{mgas}} = c_{\text{vgas}} \times \rho_{\text{gas}} \quad [2]$$

c_{vgas} = volumetric concentration [ppm]

with c_{mgas} = percentage by mass in [mg/m³]

ρ_{gas} = density of analyzed material [kg/m³]

Warning! Observe the units with the volumetric concentration and density!

$p_{\text{gas}} = \frac{M_{\text{rgas}}}{V_{\text{mgas}}} \quad [\text{kg/m}^3 \text{ or g/l}]$ with M_{rgas} = molar mass of the analyzed material [g/mol] V_{mgas} = molar volume of the analyzed material [l/mol] p_{gas} = density of the analyzed material [g/l or kg/m³]	[3]
Quelle: ISO 8178-1: 2006(E)	

Table 2: Conversion specification

For the molar volume of gaseous exhaust gases from the gas engines, the first approximation can assume a molar volume of 22.414 m³/kmol. The exact values can be obtained from the corresponding tables – extract in Appendix 2.2.

1.2.2 Emissions with reference values

Depending on the respective emission specifications, a reference value can also be specified to facilitate comparability between the different pollution emitters.

In the case of TA-Luft and the 40 CFR Part 60, Subpart JJJJ, the associated reference value is the residual oxygen content (O₂ value in Vol %) in the exhaust gas.

The conversion of the raw measured values to values with reference value is carried out according to the following specification:

$E_B = \frac{21 - O_B}{21 - O_M} \times E_M$	[4]
E _B = mass concentration related to reference oxygen content [kg/m ³]	
E _{BM} = mass concentration [kg/m ³]	
O _M = measured oxygen content [%]	
O _B = reference oxygen content [%]	
Extract from TA-Luft 2002	

Table 3: Conversion

1.2.3 Specific emissions

Specific emissions generally refer to emissions related to generated work of the engine or genset.

Specific emissions [$\frac{\text{g}}{\text{kWh}}$] = $\frac{\text{Emissions [ppm] x density x volume flow}}{[\text{kWh}]}$ [5]
Definition of specific emissions

Table 4: Specific emissions

These specific emissions can be calculated for all emissions created at the gas engine. For the calculation, we refer you at this point to the currently valid edition of ISO 8178-1 .

Special features in the EPA area:

The 40 CFR Part 60, Subpart JJJJ (cf. section 1) gives the operating company the choice of verifying the emission conformity for emissions NO_x, CO and VOCs as a limit value either with a specific emission limit, or an emission limit value with the reference value residual oxygen.

When doing this, note that for engines with a high efficiency the specific emission limit value is basically more favorable. This is because it permits leaner engine operation, e.g. based on the raw measured values related to nitric oxide, and thus permits a higher efficiency of the whole plant.

The disadvantage is that the calculation of the specific emissions as per 40 CFR Part 60, Subpart JJJJ requires the exhaust gas volume flow, the determination of which is, in part, subject to higher error tolerance.

(d) To determine compliance with the NO_x mass per unit output emission limitation, convert the concentration of NO_x in the engine exhaust using Equation 1 of this section:

$$ER = \frac{C_d \times 1.912 \times 10^{-3} \times Q \times T}{HP - hr} \quad (\text{Eq. 1})$$

[view or download PDF](#)

Where:

ER = Emission Rate of NO_x in g/HP-hr.

C_d = Measured NO_x concentration in parts per million by volume (ppmv).

1.912 x 10⁻³ = conversion constant for ppm NO_x to grams per standard cubic meter at 20 degrees Celsius.

Q = Stack gas volumetric flow rate, in standard cubic meter per hour, dry basis.

T = Time of test run, in hours.

HP-hr = Brake work of the engine, horsepower-hour (HP-hr).

Extract from 40 CFR Part 60, Subpart JJJJ – example of specific NO_x emissions

For the measurement of emissions in the EPA area („Emissions Compliance Test“), there are defined measurement methods. They can be viewed on the homepage of the EPA – Emission Measurement Center (cf. Source citations, section 7).

NOTE:

For the specific emissions as per 40 CFR Part 60, Subpart JJJJ , the mechanical engine power (el. power plus generator efficiency / gearbox efficiency) must be drawn on – cf. technical data sheet.

1.3 Gas engine emissions

The main pollutants created during gas-engine combustion are nitric oxide (NO_x), carbon monoxide and carbon dioxide as well as unburned hydrocarbons.

a. Nitric oxide - NO_x

Nitric oxides are generated during the combustion with atmospheric oxygen mainly due to the high temperatures in the combustion chamber (thermal NO_x). The term nitric oxide (NO_x) generally refers to the sum of nitrogen monoxide (NO) and nitrogen dioxide (NO_2).

b. Carbon monoxide, carbon dioxide - CO , CO_2

Both of these pollutants are created during the combustion of carbon-based energy sources, whereby carbon monoxide results from an incomplete combustion. The reduction of CO emissions in the gas engine is usually achieved by the oxidation catalyst.

c. Hydrocarbons

In the case of unburned hydrocarbons, special attention must be given to their definition in the respective exhaust gas legislation. It is possible that either the complete hydrocarbon emissions are limited, or only individual compounds (e.g. formaldehyde), or that individual compounds are possibly excluded from the summation.

Special features in the EPA area:

In the 40 CFR Part 60, Subpart JJJJ, the VOC`s (volatile organic components) are limited. In contrast to the THC`s (total hydrocarbons) measured at the test bench, the VOC limit value does not take methane, ethane or formaldehyde into consideration.